

Arnav Ajay

Houston, TX | 346-236-4067 | arnav.ajay@duke.edu | <https://www.linkedin.com/in/arnav-ajay04/>

EDUCATION

Duke University

Aug. 2022 – May 2026

BS in Computer Science and BSE in Electrical and Computer Engineering, GPA: 3.77

Durham, NC

EXPERIENCE

Robotics Engineering Intern

May 2024 – August 2024

Stryker Corporation

Weston, FL

- Developed relative positioning devices and software to perform point pair transformations, conducting statistical analyses to assess accuracy and repeatability of the positioning methods
- Utilized SolidWorks to design and simulate custom parts for medical-related relative positioning
- Leveraged MATLAB and Python to analyze positioning data, and conduct a Gage R&R analysis of the measurement systems. Translational and rotational errors were under 2mm and 2 degrees respectively
- Results produced were within goal spec value and cadaver testing for the positioning method was approved

iOS App Development Intern

June 2023 – Present

Fyzical Therapy & Balance

Durham, NC

- Working under Professor Rahib Younes to develop and deliver 3 unique iOS physical therapy apps
- Designed mobile exercises with Swift for breath training and Phantom Limb Syndrome rehabilitation
- Implementing Apple Vision and ARKit libraries in SwiftUI to create physical therapy exercises with live gaze-tracking and pupil detection features to assist Oculomotor Disorder patients

Electrical Engineering Contractor

Sep. 2022 – Aug. 2023

Durham Museum of Life and Science

Durham, NC

- Collaboratively designed a fully custom humidification system for the museum's Leafcutter Ant exhibit
- Implemented a PID control of the exhibit's humidity via an ESP32, and hosted a local web server with live updates on the humidifier status. Goal of sustained 80% humidity was achieved
- Designed 3D printed housings in SolidWorks, and used Altium to produce a custom PCB for the product

Embedded Systems Intern

June 2021 – Nov. 2021

Texas Advanced Cyber Technology

Cypress, TX

- Worked under the guidance of CEO Scott Harsany to develop elderly care and safety systems
- Devised a safety and security system via a modified Raspberry Pi hosting an array of ESP32s and sensor modules
- Tailored sensor analytics using Grafana, Firebase, and other web-based services managed by Docker containers

PROJECTS

Duke Aero Payload | ESP32 Pro S3, Raspberry Pi, Altium, KiCad, C++

Aug. 2024 – Present

- Tasked with design of the flight computer for the rocket payload, which oversees live telemetry transmission, power distribution, and communication across all subsystems
- Developing a system to capture live digital video for onboard computer vision purposes while transmitting analog video to the ground station post-deployment

Automating Medical PreAuthorization | Python, Flask, HTML/CSS/JavaScript, LLMs

May 2023 – Present

- Retrained an LLM on custom data for streamlining the PreAuthorization approval process
- User profiles are created and cross-checked with insurance policies to verify coverage for specific treatments
- Developed a full-stack web application with Flask serving a REST API with HTML/CSS as the frontend

Multimodal Machine Learning Research | TensorFlow, Keras, PyTorch, VQA

Sept 2021 – May 2022

- Worked under Professor Farshad Firouzi to build a Multimodal model for chest X-ray VQA tasks
- Constructed a multimodal learning framework using a combined vision-language encoder with self-supervised pre-training for this task (training accuracy over 90%)

SKILLS

Languages: Java, Python, C/C++, Swift, SQL (SQLAlchemy), JavaScript, HTML/CSS, MATLAB, Verilog

Software Frameworks: Flask, Django, REST API, PyTorch, TensorFlow, Xcode, Docker, Firebase, Android Studio

Engineering Skills: CAD (AutoCAD, Fusion360, SolidWorks), PSpice, MATLAB, Simulink, PCB Design (Altium, KiCad), Product Design and Management, Microcontrollers(ESP32, Arduino, Raspberry Pi), FPGA programming